

Definition 2.6 ([23, Section 4.3]). Fix a strongly E^* -unitary inverse semigroup (S, φ) with group G and morphism $\varphi : S^\times \rightarrow G$. For all $g \in G$, define the sets $E_g = \{x \in E : x \leq ss^* \text{ for some } s \in S \text{ with } \varphi(s) = g\} \subseteq E$ which are all semilattices as well. We define morphisms $\phi_g : E_{g^{-1}} \rightarrow E_g$ and that for $x \in E_{g^{-1}}$ map to $\phi_g(x) = sxs^*$ where $s \in S$ is any element where $\varphi(s) = g$ and $x \leq s^*s$. The inverse semigroup (S, φ) induces a partial action of G on the generalized Boolean algebra $\mathcal{T}_c(E)$ with ideals identified with $\mathcal{T}_c(E_g)$ and actions on ideals $\tilde{\phi}_g : \mathcal{T}_c(E_{g^{-1}}) \rightarrow \mathcal{T}_c(E_g)$ generated by $V_x \mapsto V_{\phi_g(x)}$. We denote $\Phi = (\{\mathcal{T}_c(E_g)\}_{g \in G}, \{\tilde{\phi}_g\}_{g \in G})$ define $L_R(S, \varphi) = \text{Lc}(R, \mathcal{T}_c(E)) \rtimes_\Phi G$. The R -algebra $L_R(S, \varphi)$ is generated by elements of the form $\{x\delta_g\}_{g \in G, x \in E_g}$ and calculations involving these elements are described in [23, Lemma 4.15].

As long as S is strongly E^* -unitary, the resulting algebra $L_R(S, \varphi)$ is isomorphic irrespective of the chosen group G and map $\varphi : S^\times \rightarrow G$, hence we unambiguously write $L_R(S)$ whenever we don't need to keep a specific grading in mind. For two strongly E^* -unitary inverse semigroups S_1 and S_2 , we say that $S_1 \subseteq_c S_2$ if $S_1 \subseteq S_2$ and $E_1 \subseteq E_2$ preserves finite covers. If $S_1 \subseteq_c S_2$, then for any group G and grading $\varphi : S_2^\times \rightarrow G$ on S_2 there is an induced grading $\varphi|_{S_1^\times}$ on S_1 . Furthermore, the induced partial actions from these gradings satisfy $(\mathcal{T}_c(E_1), \Phi_1) \subseteq (\mathcal{T}_c(E_2), \Phi_2)$ and hence $L_R(S_1) \subseteq L_R(S_2)$.

3. MORITA EQUIVALENCE OF SUBRINGS FOR RINGS WITH LOCAL UNITS

When we refer to Morita equivalence in this paper, we are referring to the notion of Morita equivalence in rings with local units first developed by Abrams in [4] and further explored by Anh and Marki in [24]. We will denote Morita equivalent rings with local units R and S as $R \sim_M S$.

In this section, we present a technique to prove that a subring $S \subseteq R$ of a ring with local units is Morita equivalent. We will later apply this to the partial skew group rings and derive various conditions to make the rings Morita equivalent. Most of this section is a generalization of the ideas presented in [3] and [10] to prove a desingularization result for graph and ultragraph algebras respectively.

A significant generalization of this section is removing the countability assumption used previously by being slightly more careful with our directed limits. This lets our results apply to all rings with local units instead of just rings with σ -units, which will be useful later when we apply our results to objects that aren't necessarily countable.

Definition 3.1 ([24, Definition 1]). A ring R (not necessarily unital) is called a *ring with local units* if there exists some set $E \subseteq R$ such that elements of E are idempotents, commute with each other, and $\bigcup_{e \in E} eRe = R$.

For a ring R with local units, the category of (left) R -modules is defined to be the modules M , with the additional requirement $RM = M$. These are often called unitary modules. Morphisms of R -modules are defined in the usual way.

Rings with local units R and S are said to be Morita equivalent if their respective categories $R\text{-Mod}$ and $S\text{-Mod}$ of (unitary) modules are equivalent.

All modules we consider in this section will be left modules, so we omit this from our notation.

Example 3.2. For a ring R with local units, we present examples of R -modules and morphisms that are used throughout this section.